

Java Stream Map<K,V> Interview Questions with Examples

1. Convert List to Map

Question: Convert a `List<String>` to `Map<String, Integer>` where value is string length.

Example:

```
List<String> list = Arrays.asList("apple", "banana", "cherry");
Map<String, Integer> map = list.stream()
    .collect(Collectors.toMap(
        s -> s,
        s -> s.length()
    ));
```

Output: {apple=5, banana=6, cherry=6}

2. Handle Duplicate Keys in Map

Question: How to handle duplicate keys while collecting to map?

Example:

```
List<String> list = Arrays.asList("a", "bb", "a", "ccc");
Map<String, Integer> map = list.stream()
    .collect(Collectors.toMap(
        s -> s,
        s -> s.length(),
        (v1, v2) -> v1
    ));
```

Output: {a=1, bb=2, ccc=3}

3. Group By First Character

Question: Group strings by their first character.

Example:

```
List<String> list = Arrays.asList("apple", "avocado", "banana", "berry");
Map<Character, List<String>> grouped = list.stream()
    .collect(Collectors.groupingBy(s -> s.charAt(0)));
```

Output: {a=[apple, avocado], b=[banana, berry]}

4. Convert Map to List of Keys and Values

Question: Extract keys and values from a map.

Example:

```
Map<String, Integer> map = Map.of("a", 1, "b", 2, "c", 3);
List<String> keys = map.keySet().stream().collect(Collectors.toList());
List<Integer> values = map.values().stream().collect(Collectors.toList());
```

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Output: keys = [a, b, c], values = [1, 2, 3]

5. Sort Map by Values

Question: Sort a Map<String, Integer> by values.

Example:

```
Map<String, Integer> map = Map.of("a", 3, "b", 1, "c", 2);
LinkedHashMap<String, Integer> sorted = map.entrySet().stream()
    .sorted(Map.Entry.comparingByValue())
    .collect(Collectors.toMap(
        Map.Entry::getKey,
        Map.Entry::getValue,
        (e1, e2) -> e1,
        LinkedHashMap::new
    ));
```

Output: {b=1, c=2, a=3}

6. Merge Two Maps

Question: Merge two maps using streams.

Example:

```
Map<String, Integer> map1 = Map.of("a", 1, "b", 2);
Map<String, Integer> map2 = Map.of("b", 3, "c", 4);
Map<String, Integer> merged = Stream.concat(map1.entrySet().stream(),
    map2.entrySet().stream())
    .collect(Collectors.toMap(
        Map.Entry::getKey,
        Map.Entry::getValue,
        (v1, v2) -> v1 + v2
    ));
```

Output: {a=1, b=5, c=4}